2024 IEEE International Conference on Computing, Applications and Systems (COMPAS),
September 25-26, 2024, Bangladesh

# Unleashing Machine Learning for Hepatitis C Prediction: A Holistic Exploration of Clinical Insights

Al-Amain
Dept. of Computer Science and Engineering
Southeast University
Dhaka, Bangladesh
alamainali49@gmail.com

Fatema Tuj Janin
Dept. of Computer Science and Engineering
Southeast University
Dhaka, Bangladesh
jeninahmed20@gmail.com

Faizul Ahmed Robin
Dept. of Information and Electrical Engineering
Hamburg University of Applied Sciences
Hamburg, Germany
faizulahmed.robin@haw-hamburg.de

Shifat Ahmed
Dept. of Computer Science and Engineering
Southeast University
Dhaka, Bangladesh
shifat.ahmed@seu.edu.bd

Khandaker Mohammad Mohi Uddin
Dept. of Computer Science and Engineering
Southeast University
Dhaka, Bangladesh
jilanicsejnu@gmail.com

2024 IEEE International Conference on Computing, Applications and Systems (COMPAS) | 979-8-3315-2976-5/24/$31.00 ©2024 IEEE | DOI: 10.1109/COMPAS60761.2024.10796087

***Abstract*— Chronic diseases, characterized by long-term effects and persistent symptoms, pose significant challenges to human health and well-being. The hepatitis C virus (HCV) is the cause of chronic hepatitis C, which affects 58 million people globally. It also causes 290,000 deaths annually due to liver cancer and cirrhosis, two conditions connected to HCV. Taking advantage of machine learning (ML) algorithms' ability to analyze large datasets and identify patterns and relationships that aid in diagnosis, this effort focuses on employing algorithms for machine learning to identify and forecast hepatitis C precisely. Various techniques were utilized, including Random Forest, Cat Boost, SVM, and others to create predictive models using blood values from 615 patients, including both hepatitis patients and healthy blood donors. The dataset underwent comprehensive preprocessing, including handling missing values, scaling, and dimensionality reduction via Principal Component Analysis (PCA). Various ML algorithms were evaluated, and the Cat Boost classifier achieved the highest accuracy of 99.17%, recall of 92.00%, precision of 100.00% and F1-score is 96.00%. This study demonstrates the potential of ML in enhancing HCV diagnosis, offering less invasive and cost-effective alternatives to traditional methods, and providing predictive insights for personalized treatment plans. The findings underscore the importance of advanced data preprocessing and feature selection in improving model efficiency. This study adds to the ongoing efforts to lessen the prevalence of hepatitis C worldwide, aligning with the World Health Organization's goal to significantly decrease new HCV cases and related deaths by 2030.**

***Keywords—Hepatitis C Virus Prediction, Machine Learning, PCA, Cat Boost.***

## I. INTRODUCTION

Significant obstacles are presented by chronic diseases, which are defined by enduring symptoms or long-term repercussions [1].Given that, these illnesses have a negative effect on human existence. Organizations, governmental bodies and people are actively helping those who are impacted by these conditions [2,3]. One such chronic illness is hepatitis C, which is caused by the viral infection known as hepatitis C (HCV) [4,5]. The globe's HCV infection rate is between 120 and 130 million [6], as of 2019, Additionally, there would be 58 million cases of chronic hepatitis C across the globe. Furthermore, hepatitis C-related diseases claim the lives of 290,000 individuals annually; these diseases mostly cause liver cancer and cirrhosis, which together afflict around 1.5 million people [7]. The majority of cases of this virus are discovered in developing and poor nations in Asia and Africa, as opposed to wealthy nations in Europe and North America. Chronic illnesses strike a greater percentage of people across nations like Egypt, China, and Pakistan [8,9] with high rates of chronic illnesses reported in Egypt [10,11]. Healthcare professionals (HCWs) are concerned about frequent biological risks, and personal contact raises the risk of catching HCV and other blood-borne viruses [12]. HCV therapy has improved with the introduction of direct-acting antiviral (DAA) drugs in 2014, with a therapeutic success of 95% or higher [13]. By 2030, the WHO seeks to reduce new cases of hepatitis by 90% and hepatitis-related fatalities by 65%. HCV is the most prevalent form of chronic liver disease worldwide [14,15]. Techniques of Machine Learning modes like ANN, SVM, KNN, and Logistic Regression have been widely employed in diagnosing medical conditions, including HCV. By examining enormous datasets, machine learning algorithms can uncover patterns and links for effective diagnostics. Additionally, ML algorithms can predict treatment results and create customized regimens, increasing the effectiveness and financial viability of HCV therapy. Numerous studies have demonstrated the success of ML algorithms in HCV detection. Despite the need for large labeled datasets for perfect accuracy, ML is crucial for advancing healthcare. Various ML approaches, including Random Forest, Cat Boosting, and ANN, can estimate HCV risk for early prevention. However, classification and ML techniques alone may not fully solve the problem. Building a solid information-

979-8-3315-2976-5/24/$31.00 ©2024 IEEE

Authorized licensed use limited to: Southeast University. Downloaded on October 18,2026 at 04:52:15 UTC from IEEE Xplore. Restrictions apply.

gathering model and meeting all prerequisites and criteria are essential. Additional procedures including information filtration, preprocessing, feature selection, and feature extraction are frequently needed to improve algorithm performance. In this research, these steps preceded classification, leading to greater success rates. Notably, an accuracy score of 99.16% with Cat Boost classifier was achieved.

## II. Literature REview

The verge of using the findings from groundbreaking studies on hepatitis C prediction to create precise projections. This evaluation fills in the gaps between past accomplishments and the bright future prospects. The study that has been conducted in a similar manner by several researchers is shown below. The researchers published numerous studies on hepatitis classification after they made their discoveries. Ali et al. [16] suggest a method that combines data mining, fuzzy logic, and decision trees (DT) to prevent and detect HCV infections. They made use of the 98.1% trapezoidal fuzzy number (TFN). in contrast to DT's 92.5% forecast accuracy. Abd El-Salam et al. did a laboratory analysis investigation based on machine learning on 4962 HCV patients in Egypt between 2006 and 2017 [17]. Based on data from Egypt's viral hepatitis treatment program, they used six classifiers (DT, NN, RF, NB, SVM, BN) to analyze 24 clinical characteristics from 4,962 patients with accuracies ranging from 65.6% to 68.9%, Nandipati et al. [18] provide an HCV forecasting model that includes k-nearest neighbors and random forest classifiers created especially for the application dataset available in the UCI-ML repository. Mamdouh et al. [19] constructed four machine learning models, and when tweaking the hyperparameters, they discovered that the RF model's accuracy increased to 94.88% from 94.06%. They used cirrhosis with 668 cases of mild to severe class 0 illness and class 1 cases. R and Python were used to program with different features and attributes. The components together may be used more effectively for generating superior antibody sequences that affect the HCV response [20].Yusuf Aminat Bolatito et al. used ML methods (GNB, LR, MLP, KNN, DT, SVM) on the UCI Hepatitis C dataset, achieving 87% accuracy by combining LR and MLP classifiers. [21]. Machine learning techniques are provided by Hashem et al. [22] to predict cancer of the hepatocytes in individuals suffering from lasting HCV-related liver damage. The CART method (classification and regression model) and a set of filtered input variables have been provided. These have been sorted using LR, DT, and other methods for identifying the most optimal subset of variables. For DT, CART, and LR, the accuracy values were consistently 96%, 99%, and 95.5%, respectively. Das et al. tried to categorize hepatitis using velocity-enhanced whale maximizing efficiency. Using the hepatitis dataset hosted by UCI, the researchers achieved 92.5% accuracy[23]. The research goal of John Rigg et al [24]. Machine learning algorithms have been applied in recent research to predict Hepatitis C, and the SVM model outperformed logistic regression (91%) and decision trees (88%) with the maximum accuracy of 92% [27]. aims to demonstrate the efficacy of using machine learning techniques to electronic health data in order to determine which patients should be given priority for testing for the hepatitis C virus. This will minimize the workload for medical personnel, make the best use of the resources that are available, and save health care expenditures. A sample of unlabeled patients was taken to speed up the algorithm's learning process, resulting in 16.2 million unidentified instances being included in the study's conclusions.

## III. Material and Methods

The suggested system for predicting hepatitis C consists of several processes, including EDA ( Exploratory Data Analysis ) ,feature engineering, classification, and data preparation. PCA (Principal Component Analysis ) , Scaling , Checking Missing Data and careful application of Standard Scaler are used during the data preprocessing phase. Nine different categorization algorithms are used to make an accurate hepatitis C prediction. This all-inclusive method was created especially to detect and accurately identify HCV infection cases in the dataset, which contains substantial blood values from 615 individuals. Fig. **1** provides a concise diagrammatic illustration of the technique and suggested diagnostic procedure for identifying HCV disease. This picture shows how different machine learning models are used in the training stage in order to identify which is the best. The Cat Boost Classifier proved to be the most effective model in the suggested methodology when measured by accuracy measures.

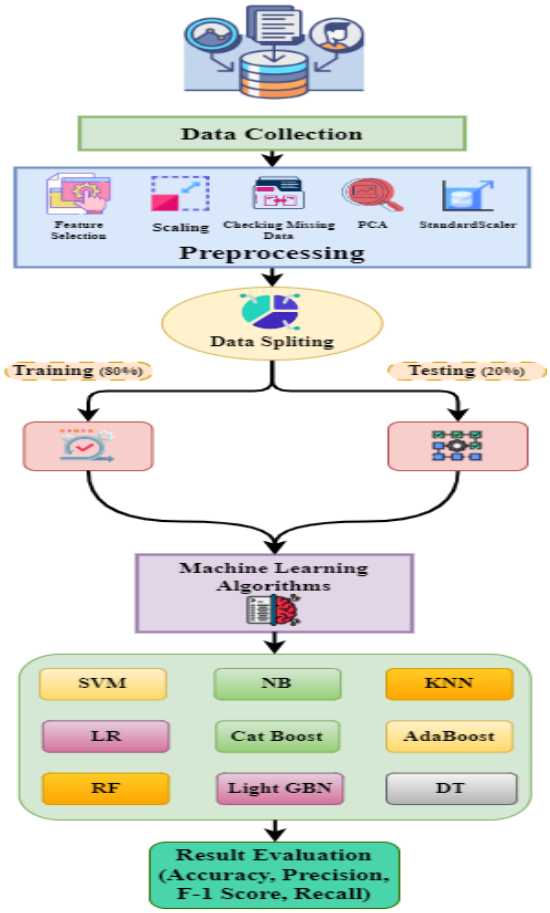



Fig 1: Working Procedure of Proposed Hepatitis C Prediction System.

Authorized licensed use limited to: Southeast University. Downloaded on October 18,2026 at 04:52:15 UTC from IEEE Xplore. Restrictions apply.

### A. Dataset

Using an online data collection tool, blood values from hepatitis patients and blood donors were recorded for the study[25] of 615 patients, 338 females and 377 men, ages ranging from 19 to 77, have been identified through this data gathering process based on certain blood characteristics, age, gender, and status of illness. On the basis of the hepatic activity index that Ishak et al. [26] established, the patients were divided into three classes. Class C1: stages F0 and F1 hepatitis without fibrosis or with just mild portal fibrosis symptoms Class C2: Fibrosis that is important to therapy (stages F2 to F6) Class C3: End-stage liver cirrhosis relevant for liver transplantation (Child-Pugh stage C). There are 615 people in total in the data collection, 377 of whom are men and 238 of whom are women. There are 139 more men than women, as can be seen. The gender distribution chart is shown in Fig 2. Sixty-one percent of people are men.

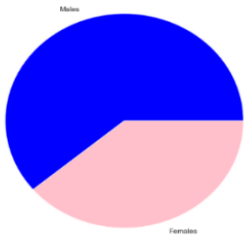



Fig 2: Gender Distribution Chart

### B. Data Preprocessing and Filtering

Initially, the dataset is imported for data manipulation and visualization. The dataset is then read into a data frame, and an overview of the data structure is obtained. Columns with unique identifiers are removed to avoid redundancy, and missing values are addressed by either dropping rows with critical missing values or imputing them with mean values for specific columns. Outliers in the dataset are capped using the interquartile range (IQR) method to mitigate their impact on model performance shown in fig.3. For machine learning algorithms to work, the categorical variables, such "Category" and "Sex," are encoded into numerical values. Next, the data is divided into testing and training sets. The characteristics undergo standardization in order to provide assurance that their means are zero and their standard deviations are one. The fig.4 displays histograms of several blood test results, illustrating how values are distributed among the different categories. Every graph shows the distribution of particular blood parameters, including age, ALB, and ALP. To further minimize the dimensionality of the data, the principal component analysis (PCA) is then carried out, and a scree plot in fig.5 is employed for displaying the explained variance. The dataset will be guaranteed to be transparent, structured, and prepared for the next steps of machine learning model training and assessment thanks to this accurate pretreatment prior to fermentation.

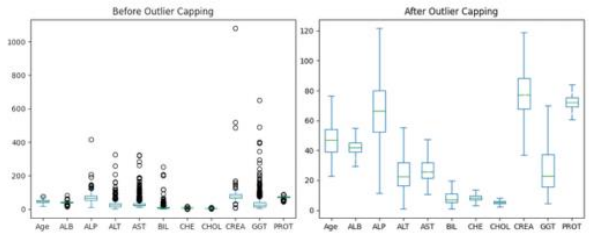



Fig 3: Box Plot of the Dataset Before and After Outlier Capping

### A. Machine Learning Models

The current study used a range of machine-learning methods to predict the occurrence of hepatitis C. They were able to do this by using information from common, reasonably priced blood tests to detect and treat patients at an early stage of the condition. Numerous machine learning methods, such as SVM, NB, KNN, Logistic Regression, AdaBoost, Cat Boosting classifier, Light GBN, RF, and DT, were carefully chosen by the researchers in order successfully achieve this aim. These algorithms were selected contingent on their efficiency in medical diagnosis.

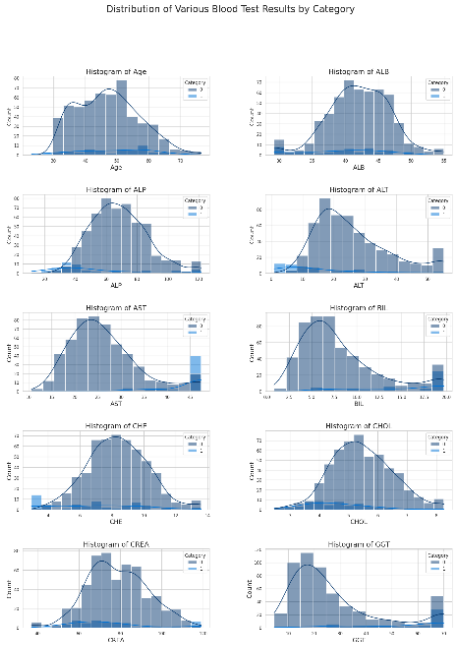



Fig 4: Histograms of Blood Test Results by Category.

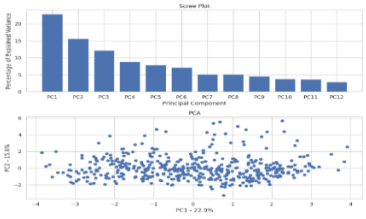



Fig 5: Principal Component Analysis: Scree Plot and PCA Scatter Plot.

Authorized licensed use limited to: Southeast University. Downloaded on October 18,2026 at 04:52:15 UTC from IEEE Xplore. Restrictions apply.

**Cat Boost:**

Cat Boost, or simply categorized boosting, is a powerful gradient-boosting technique that is applied to supervised machine learning (ML) tasks such as regression and classification. When intentional encoding is not needed, the method is particularly adept in controlling category attributes. The method builds a set of decision trees and improves estimates repeatedly across a number of boosting cycles. The Cat Boost technique integrates novel features like oblivious trees, adaptive randomly generated forests, and sequential boost in order to achieve exceptional precision and durability in real-world data. That being said, the Cat Boost approach has been acclaimed for its efficiency, user-friendliness, and its capacity to oversee intricate information. An illustration of the mathematical equation (1) that controls the Cat Boost process may be found below.

$$\boldsymbol{R(Y_i) = R_0(Y_i) + \sum_{k=1}^{K} \eta . R_k(Y_i)} \qquad \textbf{(1)}$$

## IV. Result Analysis

This study uses an 80/20 split for training and testing to use several machine learning algorithms to predict patient outcomes from a dataset. Scikit-learn and additional libraries were used to test methods such as SVM, NB, KNN, and others. In order to maximize model training, preprocessing involved addressing imbalances and abnormalities in the data. Predetermined metrics were then used to evaluate the performance of these models, aiming to improve the prediction of hepatitis C outcomes. Fig. 6 displays the correlation distribution for the HCV dataset.

### *A. Environment Setup*

The data was initial partitioned into training and testing sets utilizing randomized. In order to avoid overfitting and data leaking, the testing sets were created. The experiment's setup, which is displayed in Table 1, necessitated particular environmental conditions. All tasks have been completed on a personal computer with specs of 12.68 GB RAM, a 2-core Xeon 2.2 GHz CPU, a Tesla K80 GPU, and 71 GB of disk space using Google Co-Lab, Python tools, and other libraries.

### *B. Implementation*

There were three primary phases in the study. Prior to using machine learning techniques, the first phase required reviewing feature priority, initial processing, and information descriptions. The creation and use of several AI methods constituted the second phase. The final phase entailed visual comparisons to decide which strategy was more accurate in anticipating the spread of hepatitis C. The efficiency of each framework has been evaluated through the utilization of a confusion matrix. Python was used in Google's Co-Lab to build the visualizations and all the necessary results that were needed to fulfill our goal.

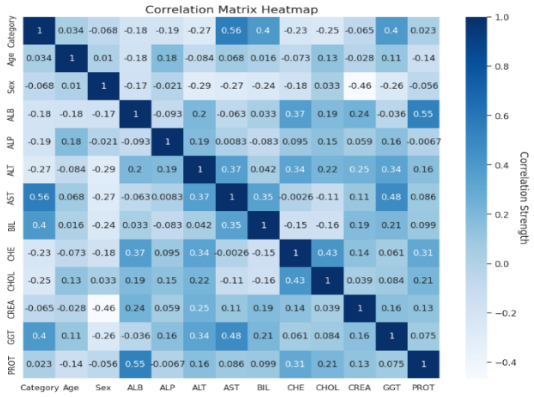



Fig 6: Heatmap of Correlation Matrix

### *C. Performance Metrics*

The effectiveness of the sorting techniques used for the HCV information categorization was assessed using a variety of measures, including recall, F1-score, accuracy, and precision. The terms false, positive, negative, and T, F, P, and N stand for unfavorable facts and the confusion matrix's judgment respectively. The total number of correctly designated healthy data points, for instance, can be expressed by TP. Equation (2) use computation to denote the overall accuracy. Recall assesses what a model can do to figure out each successful occurrence. Equation (3) provides a mathematical definition of recall. Formula (4) has been used to determine the ratio of consistently predicted favorable outcomes among all positive forecasts . formula (5) is used to determine the F1-Score metric, which assesses the harmonic mean across recall and accuracy.

$$\boldsymbol{Accuracy = \frac{TP + TN}{TP + TN + FP + FN}} \qquad \textbf{(2)}$$

$$\boldsymbol{Recall = \frac{TP}{TP + FN}} \qquad \textbf{(3)}$$

$$\boldsymbol{Precision = \frac{TP}{TP + FP}} \qquad \textbf{(4)}$$

$$\boldsymbol{F1\ Score = \frac{2 * TP}{2 * TP + FP + FN}} \qquad \textbf{(5)}$$

### *D. Result Analysis*

The precision, recall, accuracy, specificity, and F1 score of each model were ascertained using the confusion matrix, as indicated in Table 1. Below is a comparison of the effectiveness of several machine learning algorithms in identifying HCV. Higher AUC values suggest improved discriminating among the positive and negative categories in the ROC curves comparing the performance of several classifiers, as seen in Fig. 8. Table 1 provides an overview of the metrics used by several machine learning algorithms to predict the Hepatitis C virus. The accuracy, recall, precision, and the F1 score of each model are highlighted in this table, allowing for a straightforward comparison of their efficacy. With the best accuracy of 99.17%, recall of 0.92, flawless precision of 1.0, and F1-score of 0.96, the Cat Boost classifier stands out. The performance of other models, such SVM, Naive Bayes, and KNN, varies, with accuracy levels fluctuating between 92.50% to 98.34%.

Authorized licensed use limited to: Southeast University. Downloaded on October 18,2026 at 04:52:15 UTC from IEEE Xplore. Restrictions apply.

Additional information about the effectiveness of these models may be obtained by examining the matrices of confusion in Fig 7. The regression models that performed the best were Cat Boost and logistic regression. The Cat Boost classifier performed admirably in differentiating both positive and negative situations thanks to its excellent accuracy and flawless precision. Excellent outcomes were also shown using logistic regression, which had a balanced F1-score and a high degree of precision. The Random Forest and Decision Tree classifiers, on the contrary, were found to be the least effective. Notable misclassifications were identified in the Decision Tree's matrix of confusion, which decreased accuracy and F1-score. Comparably, the Random Forest classifier's poorer performance metrics were a result of its greater misclassification rates. A thorough summary of the performance of several models is given by the ROC curves displayed in Fig 8. An AUC of 1.00 was obtained by the Random Forest, Light GBM, Logistic Regression, Cat Boost, and Random Forest classifiers, indicating flawless or almost perfect performance. Moreover, the SVM classifier performed admirably, with an area under the curve of 0.99. By contrast, the AUC values of 0.92 for the Decision Tree or Naive Bayes classifiers were lower, demonstrating a discernible decline in performance. AUC values of 0.96 indicated modest performance from the KNN and AdaBoost classifiers. Overall, the analysis's findings show that the Cat Boost and Logistic Regression models perform better than the Decision Tree and Random Forest classifiers at predicting Hepatitis C. The robust predictive powers of the Random Forest, Light GBM, Cat Boost, and Logistic Regression models are further shown by the ROC curves. These results imply that Cat Boost and logistic regression are more accurate in predicting the Hepatitis C virus, especially because of their balanced performance measures and higher precision.

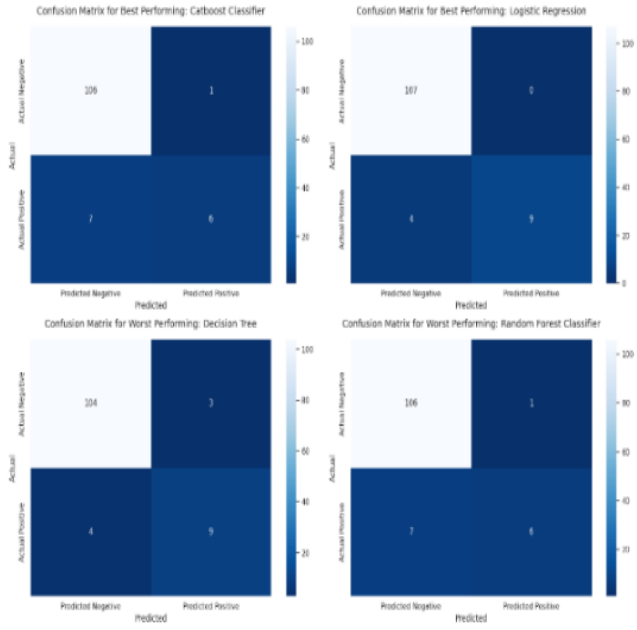



Fig 7: Comparison of Confusion Matrices: Best and Worst Performing Models

Table 1: Performance Metrics Summary of Classification Models.

| Models | Accuracy | Recall | Precision | F1-Score |
|---|---|---|---|---|
| **Cat Boost** | **99.17** | **0.92** | **1.0** | **0.96** |
| **SVM** | **98.34** | **0.61** | **0.88** | **0.72** |
| **NB** | **95.83** | **0.76** | **0.83** | **0.8** |
| **KNN** | **95.83** | **0.61** | **1.0** | **0.76** |
| **Logistic Regression** | **96.67** | **0.70** | **1.0** | **0.81** |
| **AdaBoost** | **94.17** | **0.70** | **0.75** | **0.71** |
| **Light GBN** | **95.00** | **61.54** | **88.89** | **72.73** |
| **Random Forest** | **92.50** | **0.61** | **0.88** | **0.72** |
| **DT** | **94.17** | **0.70** | **0.75** | **0.72** |

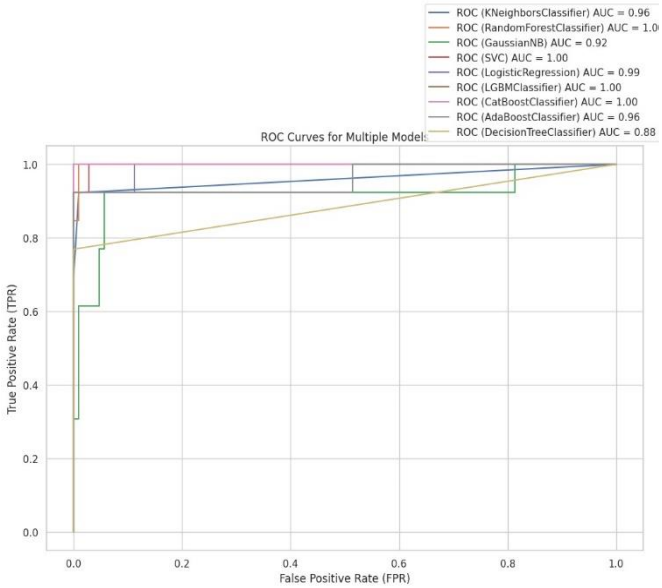



Fig 8: Performance Evaluation: ROC Analysis of Multiple Classifier.

Table 2: Comparing Several Machine Learning Algorithms to forecast hepatitis C.

| Author | Dataset | Method | Accuracy |
|---|---|---|---|
| Abd El Salam et al. [17] | Egyptian National Committee Data | RF, C4.5, MLP, NB, BN | 68.90% |
| Nandipati et al. [18] | HCV UCL-ML Repository | KNN, SVM, RF, NB, NN, Bagging, Boosting | 50.59% |
| Yusuf and Akande [21] | UCI | | 87.00% |
| Hashem et al. [22] | Al-Aini Hospital Data | LR, DT, CART | 99.00% |
| Das et al. [23] | UCI | Velocity-Enhanced whale Optimization | 92.5% |
| Anjuman Ara et al [ 27] | UCL | SVM | 94.40% |
| ***Proposed Model*** | ***UCL*** | ***Cat Boost*** | ***99.17%*** |

Authorized licensed use limited to: Southeast University. Downloaded on October 18,2026 at 04:52:15 UTC from IEEE Xplore. Restrictions apply.

## V. CONCLUSION

Notable machine learning algorithms for classification were chosen for this article's review based on classification efficacy criteria, including accuracy, recall, precision, f1 score, that hepls in finding whether a person will live or die based on the different diagnostic test findings. It has been observed during the data analysis process that the algorithms selected for the classification approach—more precisely, the algorithms for SVM, NB, KNN, Logistic Regression, AdaBoost, Cat Boost, Light GBM, RF, and DT—show some respectable accuracy percentages.. To classify the factor that depends on life or death, each of these methods should be selected above the others. The recommended method helps doctors identify HCV infections efficiently. When it comes to risk assessment and HCV infection prediction for adult patients participating in viral hepatitis screening programs, automated learning-based models could be useful. Considering that they might be verified independently Medical professionals and public health specialists should take note of these findings, which were applied to national screening programs to improve them and lower their costs. More efforts are being made to help the experts in managing and stopping the global spread of HCV by recommending a course of medication based on genetic or sequence DNA analysis algorithms. A straightforward interface also makes it feasible to classify the dataset automatically.

## REFERENCES


[1] O.O. Oladimeji, A. Oladimeji, and O. Olayanju,. Machine Learning Models for Diagnostic Classification of Hepatitis C Tests. Frontiers in Health Informatics, 10(1), p.70, 2021.

[2] J.S. Skyler, G.L. Bakris, E. Bonifacio, T. Darsow, R.H. Eckel, L. Groop, P. Groop, and Y. Handelsman, & R.E. Ratner. Differentiation of Diabetes by Pathophysiology. Natural History, and Prognosis. Diabetes, 66(2), pp.241-255, (2017).

[3] Z. Tao, A. Shi, and J. Zhao, Epidemiological perspectives of diabetes. Cell biochemistry and biophysics, 73, pp.181-185, 2015.

[4] R. Ripa, K. M. M Uddin, M. J. Alam, and M.M Rahman, Hepatitis C Prediction Using Machine Learning and Deep Learning-Based Hybrid Approach with Biomarker and Clinical Data. Biomedical Materials & Devices, pp.1-18, 2024.

[5] A.M. Al Alawi, H.H. Al Shuaili, K. Al-Naamani, Z. Al Naamani, and S.A. Al-Busafi, A Machine Learning-Based Mortality Prediction Model for Patients with Chronic Hepatitis C Infection: An Exploratory Study. Journal of Clinical Medicine, 13(10), p.2939, 2024.

[6] K. Sinka, The global burden of sexually transmitted infections. Clinics in Dermatology, 42(2), pp.110-118, 2024.

[7] C. Stasi, C., Milli, F., Voller, and C. Silvestri, The Epidemiology of Chronic Hepatitis C: Where We Are Now. Livers, 4(2), pp.172-181, 2024.

[8] A.A. Mohamed, T.A. Elbedewy, M. El-Serafy, N. El-Toukhy, W. Ahmed, and Z.A. El Din. Hepatitis C virus: A global view. World journal of hepatology, 7(26), p.2676, 2015.

[9] R. Huang, H. Rao, M. Yang, Y. Gao, J. Wang, Q. Jin, D. Ma, and L. Wei. Noninvasive measurements predict liver fibrosis well in hepatitis C virus patients after direct-acting antiviral therapy. Digestive diseases and sciences, 65, pp.1491-1500, 2020.

[10] Q. Abdulaziz Alodini. Prevalence of Hepatitis B Virus (HBV) and Hepatitis C Virus (HCV) Infections among Blood Donors at Al-Thawra Hospital Sana'a City-Yemen. Yemeni Journal for Medical Sciences, 2012.

[11] E.H. Nashaat. Lipid profile among chronic hepatitis C Egyptian patients and its levels pre and post treatment. *Nat Sci*, *8*(7), pp.83-89, 2010.

[12] C. Westermann, C. Peters, B. Lisiak, M. Lamberti, and A. Nienhaus. The prevalence of hepatitis C among healthcare workers: a systematic review and metaanalysis. Occupational and environmental medicine, 72(12), pp.880-888, 2015.

[13] O. Falade-Nwulia, C. Suarez-Cuervo, D.R. Nelson, M.W. Fried, J.B. Segal, and M.S. Sulkowski. Oral direct-acting agent therapy for hepatitis C virus infection: a systematic review. Annals of internal medicine,166(9),pp.637-648, 2017.

[14] H. Park, W.H. Lo-Ciganic, J. Huang, Y. Wu, L. Henry, J. Peter, M. Sulkowski, and D.R. Nelson. Evaluation of machine learning algorithms for predicting directacting antiviral treatment failure among patients with chronic hepatitis C infection. Scientific reports, 12(1), p.18094, 2022.

[15] J.M. Paik, P. Golabi, Y. Younossi, A. Mishra, and Z.M. Younossi. Changes in the Global Burden of Chronic Liver Diseases From 2012 to 2017: The Growing Impact of NAFLD, 2021.

[16] M.M.R. Ali, Y. Helmy, A.E. Khedr, and A. Abdo. Intelligent decision framework to explore and control infection of hepatitis C virus. In The International Conference on Advanced Machine Learning Technologies and Applications (AMLTA2018) (pp. 264- 274). Springer International Publishing, 2018.

[17] S.M. Abd El-Salam, M.M. Ezz, S. Hashem, W. Elakel, R. Salama, H. ElMakhzangy, and M. ElHefnawi. Performance of machine learning approaches on prediction of esophageal varices for Egyptian chronic hepatitis C patients. *Informatics in Medicine Unlocked*, *17*, p.100267, 2019.

[18] Nandipati, S.C., XinYing, C. and Wah, K.K., 2020. Hepatitis C virus (HCV) prediction by machine learning techniques. Applications of Modelling and Simulation, 4, pp.89-100.

[19] H. Mamdouh Farghaly, M.Y. Shams, and T. Abd El-Hafeez. Hepatitis C Virus prediction based on machine learning framework: a real-world case study in Egypt. *Knowledge and Information Systems*, *65*(6), pp.2595-2617, 2023.

[20] S. Eliyahu, O. Sharabi, S. Elmedvi, R.Timor, A. Davidovich, F. Vigneault, C. Clouser, R. Hope, A. Nimer, M. Braun, and Y.Y. Weiss. Antibody repertoire analysis of hepatitis C virus infections identifies immune signatures associated with spontaneous clearance. Frontiers in Immunology, 9, p.3004, 2018.

[21] A. YUSUF, and O. AKANDE. Hepatitis diseases prediction using machine-learning techniques. FUDMA JOURNAL OF SCIENCES, 5(3), pp.1-8, 2021.

[22] S. Hashem, M. ElHefnawi, S. Habashy, M. El-Adawy, G. Esmat, W. Elakel, A.O. Abdelazziz, M.M. Nabeel, A.H. Abdelmaksoud, T.M. Elbaz and H.I. Shousha. Machine learning prediction models for diagnosing hepatocellular carcinoma with HCV-related chronic liver disease. Computer methods and programs in biomedicine, 196, p.105551, 2020.

[23] S. Das, M. Nayak, M.R. Senapati and J. Satapathy. Medical data classification using velocity enhanced whale optimization algorithm. In 2021 First International Conference on Advances in Computing and Future Communication Technologies (ICACFCT) (pp. 18-22). IEEE, , 2021, December.

[24] J. Rigg, O. Doyle, N. McDonogh, N. Leavitt, R. Ali, A. Son and B. Kreter. Finding undiagnosed patients with hepatitis C virus: an application of machine learning to US ambulatory electronic medical records. BMJ Health & Care Informatics, 30(1) , 2023.

[25] HCV Data Data Set. UCI Machine Learning Repository. 2020. Available online: https://archive.ics.uci.edu/ml/datasets/HCV+data (accessed on 19 March 2023).

[26] G. Hoffmann, A. Bietenbeck, R. Lichtinghagen, and F. Klawonn. Using machine learning techniques to generate laboratory diagnostic pathways—a case study. Journal of Laboratory and Precision Medicine, 3(6), 2018.

[27] A. Ara, A. Sami, D.L. Michael, E. Bazgir, and P. Mandal. Hepatitis C prediction using SVM, logistic regression and decision tree. *World Journal of Advanced Research and Reviews*, *22*(2), pp.926-936, 2024.


Authorized licensed use limited to: Southeast University. Downloaded on October 18,2026 at 04:52:15 UTC from IEEE Xplore. Restrictions apply.